

Self-Correcting AI Teaching Activity Generator

An AI lesson generator that reviews and fixes its own pedagogical mistakes before a teacher ever sees them.

AI-Generated Lessons Can Quietly Fail

A single AI pass has no way to tell a teacher what it got wrong.



Unsourced or fabricated data

Contested statistics stated as fact — or worse, an invented citation that looks authoritative.



Flattened nuance

Sensitive social or historical topics reduced to a false either/or.

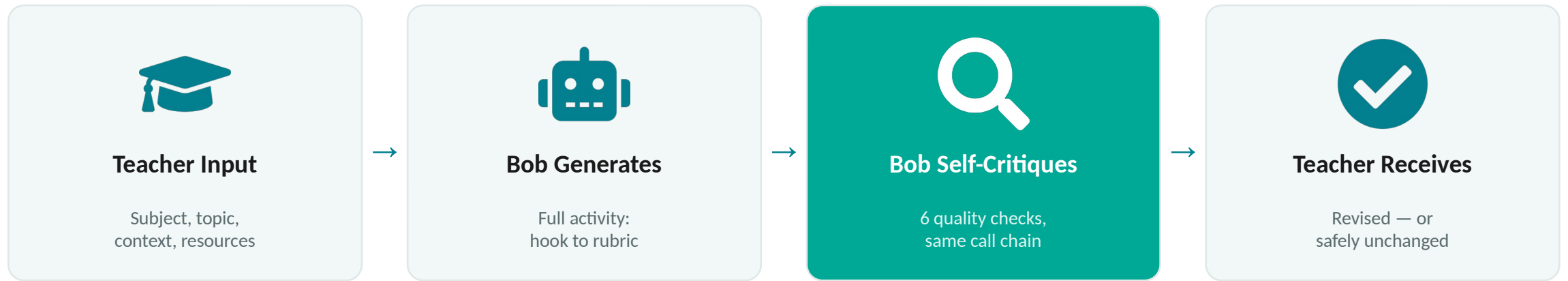


No accommodation

Every student forced into public speaking, with no low-pressure alternative.

Generate, Then Self-Critique

A second, independent Bob call reviews the first call's own output before it reaches the teacher.



If the critique step fails for any reason, the app falls back safely to the original successfully-generated activity — the teacher is never blocked.

Six Concrete Quality Checks

Identified through a manual pedagogical audit — not generic AI safety boilerplate.



Data Sourcing

Flags unverified figures — never fabricates a source.



False Binary

Catches nuance flattened into a false either/or.



Scope vs. Timeline

Flags content too ambitious for the stated period.



Rubric Exemplars

Adds a concrete example to every grading level.



Public-Speaking Opt-Out

Ensures a low-pressure alternative is built in.



Internal Consistency

Extracts every claim and cross-checks for contradictions.

The Bug We Caught — and Fixed

When told to "add sourcing," Bob invented a fake academic citation — confirmed nonexistent via web search.



BEFORE

"Bhaskar et al., The Oxford Handbook of the Partition of India, 2023"

A real-sounding book that does not exist — verified against multiple academic sources. A teacher would trust and repeat it.



AFTER

*"estimates vary across sources due to incomplete historical records
⚠️ [TEACHER: verify before class]"*

A hardcoded rule now forbids fabricating any citation — uncertainty is flagged for the teacher, never invented.

Hands-On, Adapted to What's Available

Same concept, same rigor — materials calibrated to the classroom's real resources.



WELL-RESOURCED SCHOOL

Volume lesson generated live:
graduated cylinders, precision lab
equipment, 3D-printed test shapes.



LOW-RESOURCE SCHOOL

Same lesson, same live test:
jerricans, clay pots, string, and
containers students bring from home.

Tested Before It Shipped

Every mechanism here was iterated against real failures before being wired into the live app.

10

initial activities
manually audited

5

adversarial test
scenarios run

5

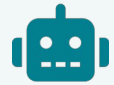
prompt iterations
(v1 → v5)

2

real bugs found
& fixed

Scenarios covered: a physics lab, a sensitive history topic (1947 Partition of India), a contested policy debate, a mock trial, and a science fair-testing lab.

IBM Bob 2.0 — Two Distinct Roles



RUNTIME ENGINE

Every teacher request spawns the Bob CLI twice: once to generate the activity, once to self-critique it — both via the same subprocess and JSON-parsing pipeline.



DEVELOPMENT-TIME AGENT

Bob CLI also wrote our standalone quality-validator script, later found and fixed a real bug in it, and made the precise, verified edits that wired the self-critique and hands-on guidance into the app itself.

Stack: Node.js + Express · IBM Bob 2.0 CLI · Anthropic Claude (prompt design, testing, independent fact-checking)



Quality Education, at Any Resource Level

Every classroom — fully equipped or improvised — deserves a lesson that was checked for real pedagogical defects before it ever reached a student.

UN SDG 4 — Quality Education

Thank you.